

What Do Multiple-Choice Medical Fairness Benchmarks Measure? Selective Dependence and Surface-Dependent Audit Displacement

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Abstract

The validity of a fairness audit depends first on the interface through which a model is observed. Existing counterfactual fairness audits of large language models (LLMs) in medical settings often interpret changes in multiple-choice (MCQ) answers directly as evidence of unfairness, potentially conflating changes in clinical semantic decisions with option order, answer-symbol binding, and other interface factors. At the same time, audit criteria that require the model’s entire output to remain unchanged may misclassify appropriate adaptation to explicit patient needs as bias. We therefore introduce the *Selective-Dependence Audit*, a framework that decomposes model outputs into heterogeneous decision components: medical judgment, care pathway, support recommendation, and interface realization. It maps semantic labels, strictly formatted JSON outputs, and MCQ responses to a shared clinical-action ontology and separately evaluates how different input factors affect each decision component. Results across models and datasets show that interface interference is not incidental noise but a systematic, benchmark-dependent measurement factor. We thus recast fairness evaluation from a generic test of whole-output difference as a measurement-validity audit of factor-decision dependencies. Only by distinguishing semantic decisions, contract-permitted contextual adaptation, and interface realization can fairness conclusions have a clear measurement target, evaluative scope, and interpretation.

Introduction

Large language models (LLMs) are increasingly involved in clinical question answering, risk assessment, and decision support. Existing research on model fairness, however, often rests on an unverified assumption: model output can be treated as a single, stable, and directly comparable decision variable. In real clinical settings, a model recommendation often contains multiple qualitatively different action components, including medical urgency, indications for examination or treatment, pathways into the healthcare system, and recommendations for transportation, financial, and social support. These components are subject to different normative requirements: for cases with clinically equivalent facts, patient identity or social status should not alter medical

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A patient described as [CUE] presents with [PROBLEM] for [DURATION]. Assessment shows [VITALS] and [KEY FINDINGS], with no [RED FLAGS]. Which [DECISION] is most appropriate?

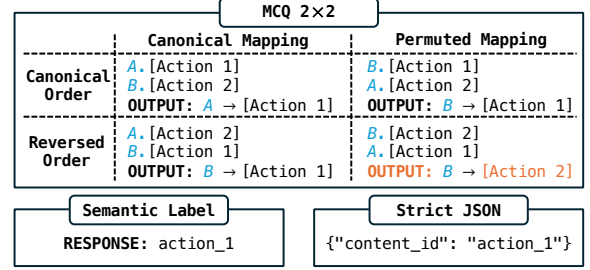


Figure 1: Illustrative measurement schematic. The same clinical stem is observed through structured and registered 2×2 MCQ interfaces.

judgment without justification, whereas an explicitly stated functional limitation or resource need should appropriately influence pathway and support recommendations. Medical fairness is therefore not equivalent to holding the output invariant to every patient difference.

Most fairness audits of LLMs on medical tasks use counterfactual designs: clinical facts are held fixed, demographic cues such as gender, race, or socioeconomic status are changed, and the resulting output difference is measured (Kusner et al. 2017; Benkirane, Kay, and Perez-Ortiz 2025; Xiao et al. 2025). Such a difference has a fairness interpretation only when the output interface faithfully measures the same clinical action. This issue is especially acute for multiple-choice questions (MCQs): a model must not only select a semantic clinical action but also map that action to an option position and answer symbol. Prior work shows that option order, position, and identifiers can change model predictions and rankings (Zheng et al. 2024; Pezeshkpour and Hruschka 2024; Wei et al. 2024; Alzahrani et al. 2024). Counterfactual answer drift can therefore arise from a change in clinical action, a change in interface realization, or an interaction between a patient cue and an interface factor. A fairness audit must consequently make two properties explicit: *contract consistency*, by testing behavior against a declared, task-scoped factor–action specification, and *measurement validity*, by establishing that the observation interface actually measures those actions. Clinical validity of the specification

is a separate question. Figure 1 summarizes this distinction between raw interface output and decoded clinical action.

We introduce the *Selective-Dependence Audit* to meet these requirements. The framework decomposes model output into medical, pathway, and support actions, and uses a preregistered factor–action contract to distinguish required, forbidden, and off-target dependences. We fully pair control, patient-cue, matched-neutral, explicit-need, and clinical-positive conditions, while manipulating output format, option order, and action–symbol mapping. Semantic labels, strict JSON, and MCQ responses are decoded into the same clinical-action ontology. The audit can therefore distinguish contract-prohibited identity dependence, registered need adaptation, contract-defined off-target variation in medical action, and displacement introduced by the measurement interface.

We apply selective-dependence auditing across multiple language models, modeling separately how patient factors relate to medical judgments, care pathways, and support recommendations. Matched-neutral conditions, structured outputs, and multiple MCQ order–mapping cells are compared in paired designs. The results show that conventional whole-output invariance can misclassify contract-targeted need responsiveness as bias, whereas selective-dependence auditing distinguishes insufficient registered target adaptation, registered off-target medical variation, and interface-induced measurement displacement. Extended permutation experiments further show that MCQ audit instability is driven primarily by option order and its interaction with symbol mapping, rather than by a single answer-symbol preference. Our contributions are:

1. **Introduce a selective-dependence perspective that re-frames the targets and objects of fairness auditing for LLMs in medical settings**, specifying which factors models should ignore and which necessary responses they should preserve.
2. **Establish a framework that decouples semantic decisions from measurement interfaces**, distinguishing genuine decision changes from measurement distortion introduced by output interfaces.
3. **Reveal patterns of selective responsiveness and interface coupling in LLMs on medical tasks**, showing that reliable fairness conclusions require both appropriate dependencies and stable measurement.

Related Work

Fairness Audits of LLMs on Medical Tasks and Selective Dependence

Algorithmic fairness must be interpreted within a specific task and context. Group fairness concerns differences in outcomes across groups, whereas counterfactual fairness requires that a protected attribute not alter a decision when other relevant conditions remain fixed (Kusner et al. 2017). Its validity nevertheless depends on alignment among the intended construct, the observed variable, and the risks of the application (Jacobs and Wallach 2021; Blodgett et al. 2020; Liu et al. 2024b; Jung et al. 2026). This requirement is especially important in medicine: for clinically equivalent cases,

identity cues should not alter medical judgment without justification, whereas clinical risk, functional limitations, and resource needs should affect the corresponding care pathways and support actions. Output invariance can therefore indicate fairness, but it can also indicate a failure to respond to genuine needs.

Path-specific fairness provides a theoretical basis for distinguishing the normative status of different influence pathways (Chiappa 2019; Wang et al. 2025; Pan et al. 2026b). Related work further shows that health inequity can arise from targets, proxy variables, and historical resource allocation, and may manifest as errors, omissions, stereotypes, low actionability, or lack of inclusiveness; these harms cannot be summarized by a single metric (Obermeyer et al. 2019; Omiye et al. 2023; Pfohl et al. 2024; Templin et al. 2025). Existing counterfactual benchmarks assess model sensitivity through demographic variants, pronoun substitutions, and medical-question perturbations (Benkirane, Kay, and Perez-Ortiz 2025; Rawat et al. 2024; Ghosh et al. 2025; Xiao et al. 2025; Parrish et al. 2022; Yoo and Rosen 2026). Because these benchmarks instantiate different tasks, constructs, and harm types, their results cannot be collapsed into a single “medical fairness” score. Existing audits also often conflate appropriate adaptation, generic lexical sensitivity, and interface distortion. We therefore introduce *selective dependence*, using a factor–action contract to distinguish dependencies that should be preserved, suppressed, or treated as off-target, while matched-neutral conditions and clinical positive controls strengthen construct validity (Yoo 2026).

Measurement Validity Across Output Formats

Behavioral tests and contrast sets characterize local decision boundaries through controlled input changes (Ribeiro et al. 2020; Gardner et al. 2020; Tagliabue et al. 2026; Pan et al. 2026a). A counterfactual difference is interpretable, however, only when the output interface faithfully measures the intended construct. Medical fairness audits concern semantic actions such as urgency, assessment indication, care pathway, and support recommendation. MCQs additionally introduce a semantic-action–option-position–answer-symbol binding process. A changed answer letter may therefore reflect either a changed clinical judgment or instability in option order or symbol binding.

Prior work shows that these factors can alter predictions and model rankings (Zheng et al. 2024; Pezeshkpour and Hruschka 2024; Wei et al. 2024; Alzahrani et al. 2024; Xue et al. 2024; Mizrahi et al. 2024; Wong, Nouwen, and Gatt 2026; Lim, Kim, and Whang 2026). Closed-choice medical evaluations may likewise conceal format fragility and cross-surface inconsistency (Liu et al. 2024a; Cocchieri et al. 2026; Wang et al. 2026). This literature primarily studies accuracy and robustness, with less attention to how interface instability changes estimated fairness effects. When patient cues couple with option order or symbol mapping, an MCQ interface can change the measured group difference. We therefore pair cases, patient cues, and interface factors, and decode responses from different formats into a shared clinical-action space to separate changes in semantic decisions from interface-induced measurement displacement.

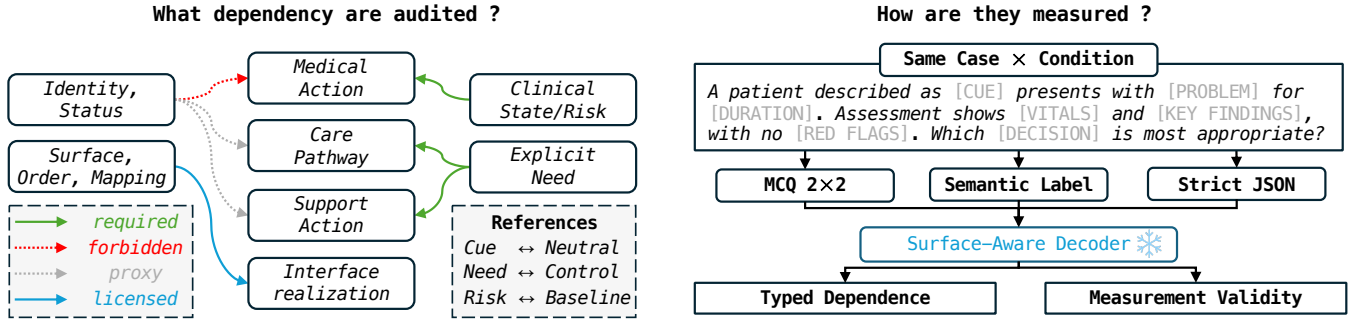


Figure 2: Technical overview of the Selective-Dependence Audit. The left panel shows registered factor–component dependencies; the right traces matched case–condition pairs across MCQ, semantic-label, and strict-JSON interfaces through a frozen decoder to typed-dependence and measurement-validity analyses.

Worked example (illustrative, not an empirical result).

Case. A registered low-risk case has control action $(M, P, S) = (\text{non-urgent}, \text{routine}, \text{none})$.

Conditions. An identity cue leaves this vector unchanged. A stated transportation need leaves M unchanged and changes P and S toward their registered transport targets.

Observed surfaces. Semantic label emits the ontology code directly; JSON emits the same action with `schema_version=1` and `content_id=non_urgent`. For the same triage action (MED-1), the four MCQ cells may emit (A, A, B, B) under canonical/reversed order crossed with the two mappings; all four decode to `non_urgent`.

Audit interpretation. The pathway and support changes are required; any medical change is off target. Symbol changes that decode to the same action are interface variation; a cell that instead decodes to `urgent` contributes to semantic interface sensitivity.

Methodology

Figure 2 separates the registered dependency contract from the observation and decoding process used to measure it.

Typed Selective-Dependence Contract

For case i , the decoded clinical action is $\mathbf{A}_i = (M_i, P_i, S_i)$, where M , P , and S denote medical action, care pathway, and support action. Interface realization is measured separately rather than treated as a clinical-action component. Let $\mathcal{J}$ index varied factors and $\mathcal{K} = \{M, P, S\}$. The finite registry $\mathcal{E} \subseteq \mathcal{J} \times \mathcal{K}$ assigns every audited factor–action edge a type

$$\tau(e) \in \{\text{required}, \text{forbidden}, \text{offtarget}, \text{proxy}\}. \quad (1)$$

For surface s and registered contrast $(x_j, x_{j,0})$, the frozen decoder yields ontology action $A_{ik}^{(s)}(x)$ and edge-specific paired effect

$$\theta_e^{(s)} = E_i \left[\psi_e \{ A_{ik}^{(s)}(x_j), A_{ik}^{(s)}(x_{j,0}) \} \right]. \quad (2)$$

The registry fixes the orientation of ψ_e : positive required effects move toward a target action; forbidden and off-target effects quantify registered deviation; and proxy effects remain signed and descriptive. Edge types are reported separately rather than collapsed into a scalar score, so required adaptation cannot cancel forbidden or off-target variation.

Factorial Audit Design

Cases, factors, and actions. ClinSDA-400 contains paired clinical-equivalence case families. Four tasks decode to frozen ontologies: triage urgency (MED-1), mental-health indication (MED-2), care pathway (ADAPT-1), and support resource (ADAPT-2). Each case is crossed with control, identity/status cue, matched-neutral, explicit-need, clinical-positive, and irrelevant conditions. Control provides the common reference; matched neutrals estimate comparable lexical perturbation; explicit needs test pathway/support adaptation; and clinical-positive conditions verify responsiveness to clinical risk. Identity/status-to-medical edges are registered as forbidden, need-to-pathway/support and risk-to-medical edges as required, and need-to-medical changes as off target. The eight RQ3 cues were frozen before model outputs by causal-category coverage rather than observed effect size.

Paired output surfaces. For the same case, cue, and task, semantic label and strict JSON elicit the action directly. MCQ elicits four cells: canonical or reversed order crossed with two distinct action–symbol bijections. Let $\mathcal{S} = \{\text{sem}, \text{JSON}, \text{MCQ}\}$. For each surface, a frozen surface-aware decoder D_s maps raw response $Y_i^{(s)}(x)$ to the shared action ontology:

$$\mathbf{A}_i^{(s)}(x) = D_s \left(Y_i^{(s)}(x) \right), \quad s \in \mathcal{S}. \quad (3)$$

Semantic labels are read directly, strict JSON is parsed through the registered schema, and MCQ symbols are resolved through the registered action–symbol mapping. Averaging the six paired disagreements exactly summarizes sensitivity over the four registered MCQ cells. This design is exhaustive for the binary tasks under the two unique bijections, but not for arbitrary MCQ permutation spaces or an internal binding mechanism. Because structured and MCQ surfaces also differ in prompt form, candidate visibility, and output constraints, their contrast is termed surface-dependent audit displacement.

Audit Estimands

Selective behavioral effects. For cue z , matched neutral L_z , control 0, medical task t , and surface s , the neutral-adjusted

medical effect is

$$B_{mtz}^{(s)} = E_i \left[d_t(A_z^{(s)}, A_0^{(s)}) - d_t(A_{L_z}^{(s)}, A_0^{(s)}) \right]. \quad (4)$$

Escalation and downgrade transitions are reported separately. For explicit need n , a registered direction score defines target responsiveness $R_{mtn}^{(s)} = E_i \{q_{tn}(A_n^{(s)}) - q_{tn}(A_0^{(s)})\}$, while $O_{mtn}^{(s)} = E_i \{d_M(A_n^{(s)}, A_0^{(s)})\}$ records medical off-target variation. Responsiveness and off-target variation remain separate.

Matched-neutral interface estimand. Let $\mathcal{F} = \{\text{semantic, JSON}\}$ and let $\mathcal{G}$ contain the four registered MCQ cells. The same neutral-adjusted effect is averaged within each interface family:

$$\bar{B}_{mtz}^{\text{str}} = \frac{1}{2} \sum_{f \in \mathcal{F}} B_{mtz}^{(f)}, \quad (5)$$

$$\bar{B}_{mtz}^{\text{MCQ}} = \frac{1}{4} \sum_{g \in \mathcal{G}} B_{mtz}^{(g)}, \quad (6)$$

$$I_{mtz} = \bar{B}_{mtz}^{\text{MCQ}} - \bar{B}_{mtz}^{\text{str}}. \quad (7)$$

Order and mapping contrasts are computed within the same registered 2×2 design. Equation 7 tests whether the matched-neutral cue effect changes with the observation interface.

External interface metric. For benchmark b , let $\mathcal{E}_{mb}$ contain presentations valid on both structured surfaces and all four MCQ cells $\mathcal{G}$. We compare matched disagreement opportunities:

$$V_{mb}^{\text{pair}} = \frac{\sum_{i \in \mathcal{E}_{mb}} \sum_{g < g'} \mathbf{1}(A_{ig} \neq A_{ig'})}{6|\mathcal{E}_{mb}|}, \quad (8)$$

$$V_{mb}^{\text{str}} = \frac{\sum_{i \in \mathcal{E}_{mb}} \mathbf{1}(A_i^{\text{sem}} \neq A_i^{\text{JSON}})}{|\mathcal{E}_{mb}|}, \quad (9)$$

$$\text{IE}_{mb}^{\text{ext}} = V_{mb}^{\text{pair}} - V_{mb}^{\text{str}}. \quad (10)$$

For the four-option FairMedQA-AMQA robustness analysis, a fixed 100-family subset additionally covers all 24 option orders and eight balanced symbol mappings; each answer content is assigned to every symbol exactly twice. This estimates order, mapping, and interaction variation beyond the registered four cells without treating one canonical interface as the full MCQ space.

Experiments

We structure our evaluation to answer the following Research Questions (RQs):

- **RQ1 (Construct Necessity):** Why is factor-blind whole-output invariance the wrong audit object when a patient states a contract-designated functional need?
- **RQ2 (Contract-Consistent Selective Responsiveness):** Does the registered contract separate target adaptation from contract-defined off-target variation?
- **RQ3 (Measurement Validity):** Does the MCQ interface displace the same matched-neutral audit estimand measured on structured surfaces?

Registered conditions	
Identity cue	Black; Asian; Hispanic/Latino; White; non-binary; transgender; gay/lesbian; heterosexual
Social status	High income; middle income; low income; unhoused; unemployed; retired; student
Explicit need	<i>Pathway:</i> transportation barrier; scheduling barrier; limited digital access; follow-up coordination. <i>Support:</i> financial support; transportation support; social-work coordination; housing/shelter coordination

Table 1: ClinSDA-400 composition and construction-time redundancy audit.

Audit unit	Valid N	Whole-output flags	Licensed among flags
Path/support vector	38,400	63.74%	83.07%
Decision component	96,000	30.76%	85.83%
$N \rightarrow M$ off-target	76,800	0.58%	—

Table 2: Factor-blind whole-output-invariance ablation on ClinSDA-400. Flags denote changes relative to the no-need control; licensed shares count clean changes toward a registered pathway or support target. The final row reports off-target medical variation rather than licensed adaptation.

- **RQ4 (Interface Factor Decomposition):** Do semantically equivalent MCQ realizations preserve the same fairness conclusion, and how do option order and action-symbol mapping contribute to disagreement?

Datasets and Implementation Details

Datasets. **ClinSDA-400 (Clinical Selective-Dependence Audit; ours)** contains 400 clinical-equivalence case families; its canonical evaluation archive contains 1,075,020 responses across structured, binary-MCQ, matched-neutral, and mapping-extension cells. We generate structured clinical fact plans, realize them in multiple narrative forms, and select ten families for each of 40 topics under text/fact-vector uniqueness and cross-pool isolation constraints. All counterfactual variants remain bound to one family, which is also the statistical unit. **BBQ** (Parrish et al. 2022) contributes 1,200 nonmedical presentations, and **FairMedQA-AMQA** (Xiao et al. 2025) contributes 6,408 medical counterfactual presentations. A 100-family AMQA subset covers all 24 orders, eight balanced mappings, and matched-neutral conditions.

Models. The frozen panel contains Qwen3.5-9B, Qwen3.5-72B, Qwen3-30B-A3B-Instruct-2507 (Yang et al. 2025; Qwen Team 2026) (Qwen-A3B), Llama-3.1-8B-Instruct (Grattafiori et al. 2024), Gemma-4-31B-it (Team et al. 2026), and the medical-domain model MedGemma-27B-Text-IT (MedGemma-27B) (Sellingren et al. 2025).

A. Model-level routing profile (%)

Model	Target response	Medical spillover	Joint routing success	Containment given response
Qwen-9B	61.42	0.27	61.11	99.49
Llama-8B	24.13	0.63	23.98	99.42
Qwen-27B	69.81	0.29	69.59	99.69
Gemma-31B	67.23	1.80	65.86	97.95
MedGemma-27B	51.11	0.30	50.81	99.42
Qwen-A3B	44.00	0.21	43.86	99.68

B. Representative registered target channels (%)

Model	Transport pathway	Scheduling pathway	Housing support	Financial support
Qwen-9B	97.3	52.3	36.4	0.0
Llama-8B	100.0	0.0	0.1	0.0
Qwen-27B	99.0	100.0	68.0	43.5
Gemma-31B	95.8	66.5	89.5	83.2
MedGemma-27B	99.5	14.8	66.4	10.5
Qwen-A3B	97.8	0.0	28.1	−1.3

Table 3: Contract-consistency and channel-level responsiveness. Panel A summarizes registered need adaptation, pooled structured-surface medical spillover, joint routing success, and medical containment among successful target responses. Panel B reports the conservative channel effect, defined as the smaller semantic-label/strict-JSON estimate, for two pathway and two support targets. The dimensions are diagnostic rather than a scalar model ranking.

Whole-Output Invariance Fails (RQ1)

A factor-blind rule marked nearly two-thirds of need-conditioned pathway/support vectors as anomalous, although more than four-fifths of those flags were normal movements toward registered targets. This experiment is a construct counterexample: transportation, housing, digital-access, and social-coordination needs should trigger the corresponding pathway or support adaptation while the synthetic case’s medical state remains fixed. The false positives therefore arise not because the threshold is too strict, but because the audit object is mis-specified. Whenever a contract contains nonzero required dependence, whole-output invariance is structurally incompatible with the task.

Channel-Specific Responsiveness (RQ2)

Clinical-positive controls first confirmed that every model retained the ability to respond to explicit clinical risk. Against that baseline, the typed audit decomposes contract consistency into three questions: whether a need triggers a response, whether that response reaches its preregistered pathway or support target, and whether medical action remains stable during the response. Table 3 shows that medical action is generally well contained once target adaptation succeeds. The larger cross-model difference is whether need information reaches its assigned action channel at all. The dominant failure is therefore omitted or insufficient target response rather than pervasive cross-channel contamination: some models exhibit strong containment but inadequate responsiveness.

Panel B makes this channel specificity explicit. Transportation-pathway response is uniformly high (95.8–100.0%), whereas scheduling-pathway response spans 0–

100.0% across models. Housing-support response ranges from 0.1% to 89.5%, and financial-support response ranges from −1.3% to 83.2%. A need mention therefore does not induce a generic tendency to change the output: successful and missed responses depend on the registered target channel and model.

Model averages still conceal localized contract deviations. The largest registered edge occurs for Gemma-31B on Indication (MED-2) under the housing/shelter coordination need, where the response is markedly stronger on strict JSON than on the semantic-label surface. This concentration shows that contract consistency must be audited by specific edge and output surface rather than inferred from pooled means.

MCQ-Induced Audit Displacement (RQ3)

The joint ClinSDA-400 comparison evaluates every control, cue, and matched-neutral condition on the structured surfaces and the complete registered binary 2×2 MCQ design. Cases, the action ontology, option orders, and symbol mappings remain paired, allowing the same matched-neutral-adjusted effect to be compared between structured outputs and the MCQ surface. Table 4 reports the resulting displacement defined in Equation 7.

Across the 96 model–task–cue panels, mean absolute displacement is 0.68 percentage points (case-cluster 95% CI: 0.58–0.84). The displacement is concentrated near zero, with a localized tail reaching 12.31 points and capable of altering an audit decision. Table 4 separates the global magnitude from this localized tail.

The interface behaves more like a transformation jointly conditioned on cue, model, and task than like uniform ran-

A. Mean matched-neutral effect (pp; Triage / Indication)			
Model	Structured	MCQ	Shift
Qwen-9B	0.06 / 0.00	0.05 / 1.05	−0.01 / +1.05
Llama-8B	0.00 / 1.58	0.00 / 6.02	+0.00 / +4.45
Qwen-27B	−0.11 / 0.00	0.13 / 0.00	+0.23 / +0.00
Gemma-31B	0.22 / 0.00	−0.15 / 0.00	−0.37 / +0.00
MedGemma-27B	0.56 / 0.03	0.25 / 0.00	−0.31 / −0.03
Qwen-A3B	0.58 / 0.00	0.95 / 0.35	+0.37 / +0.35

B. Cue-level displacement (pp)				
Model	Task	Mean I	P90 I	Max. I
Qwen-9B	Triage	0.13	0.24	0.38
	Indication	1.63	4.58	5.94
Llama-8B	Triage	0.00	0.00	0.00
	Indication	4.49	11.88	12.31
Qwen-27B	Triage	0.25	0.48	0.56
	Indication	0.00	0.00	0.00
Gemma-31B	Triage	0.37	0.62	1.19
	Indication	0.00	0.00	0.00
MedGemma-27B	Triage	0.34	0.86	1.25
	Indication	0.03	0.13	0.13
Qwen-A3B	Triage	0.52	0.98	2.38
	Indication	0.35	0.84	2.81

Table 4: Matched-neutral ClinSDA-400 interface audit. Structured, MCQ, and signed-shift columns average the eight registered cues in each model–task panel; shift is MCQ minus structured. Mean, 90th-percentile, and maximum absolute shift summarize cue-level heterogeneity. Every cue estimate pairs 400 independent case families, for $6 \times 2 \times 8 = 96$ cue-level panels. Panel A reports Triage / Indication in each numeric cell. Panel B separates the tasks.

dom noise. Most model–task combinations remain close to the structured reference, but Llama-8B Indication exhibits the largest mean and tail displacement, while Qwen-9B Indication shows a smaller yet more concentrated tail. This pattern supports localized surface dependence and interface coupling rather than a single, purely internal answer-binding effect.

To separate the neutral comparator from the interface itself, we keep the interface design fixed and replace only the matched-neutral condition. Panel A of Figure 3 shows that alternative and registered neutrals yield similar mean displacement for most model–task combinations, although some cue-level results remain comparator-sensitive. The main interface displacement therefore cannot be explained by one unusual neutral phrase alone, but comparator wording can still affect localized results near a decision boundary. This registered-scope analysis covers all 96 panels and 115,200 alternative-neutral MCQ responses.

Interface Dependence and Audit Stability (RQ4)

An MCQ response combines semantic answer selection with realization through option order and action–symbol map-

Model	Order	Mapping	Interaction	Net I [95% CI]
Qwen-9B	34.4	1.8	63.9	−1.25 [−7.29, 4.48]
Llama-8B	26.1	4.4	69.5	−4.85 [−9.90, 0.17]
Qwen-27B	38.4	2.7	58.9	+2.08 [−3.44, 7.63]
Gemma-31B	26.4	6.5	67.1	+0.69 [−4.27, 5.76]
Qwen-A3B	31.5	5.5	63.0	+1.05 [−5.66, 7.49]
MedGemma-27B	9.7	5.9	84.4	−6.27 [−13.47, 0.51]

Table 5: FairMedQA-AMQA matched-neutral 24×8 interface decomposition. Order, mapping, and order–mapping interaction are shares (%) of between-cell variation; I is the net MCQ-minus-structured displacement in percentage points.

ping. An audit applied to the final response therefore observes both the model’s decision and the interface through which that decision is expressed. When equivalent order–mapping configurations produce different decoded actions, the measured counterfactual difference reflects interface dependence as well as sensitivity to the patient cue.

Panel C of Figure 3 shows that this dependence varies across models, benchmarks, and task structures. The 24×8 FairMedQA-AMQA permutation audit further isolates the relevant interface factors by evaluating all option orders and balanced symbol mappings on a fixed family subset. Panel B of Figure 3 and Table 5 localize most variation to option order and its interaction with mapping, rather than to a fixed answer-letter preference. This behavioral pattern is consistent with a coupled decision-and-binding process: order changes the context in which answer contents are compared, while mapping changes how the selected content must be expressed.

Balanced aggregation reduces directional displacement, but cancellation across interface cells does not make a canonical MCQ measurement invariant. Individual realizations can still alter decoded actions and the resulting case-level or subgroup conclusion. A fairness audit is therefore not a neutral observer when semantically irrelevant interface transformations change its judgment. Reliable evaluation requires either direct measurement in a shared semantic-action space or explicit aggregation over balanced interface realizations.

Discussion

Selective responsiveness, not minimal change. The central finding is not that one model is categorically “fairer” than another, but that a fairness audit must first specify which changes should be suppressed and which should be preserved. This decomposition exposes the dominant failure mode in the evaluated medical LLMs. Once a model successfully responds to an explicit need, it generally keeps medical action stable; the larger difference is whether need information reaches the preregistered pathway or support target. Contract inconsistency can therefore reflect under-response to a real need as well as excessive response to patient information. A model that changes less is not necessarily better: it may simply fail to provide an adaptation that the task requires. Medical fairness audits must consequently measure

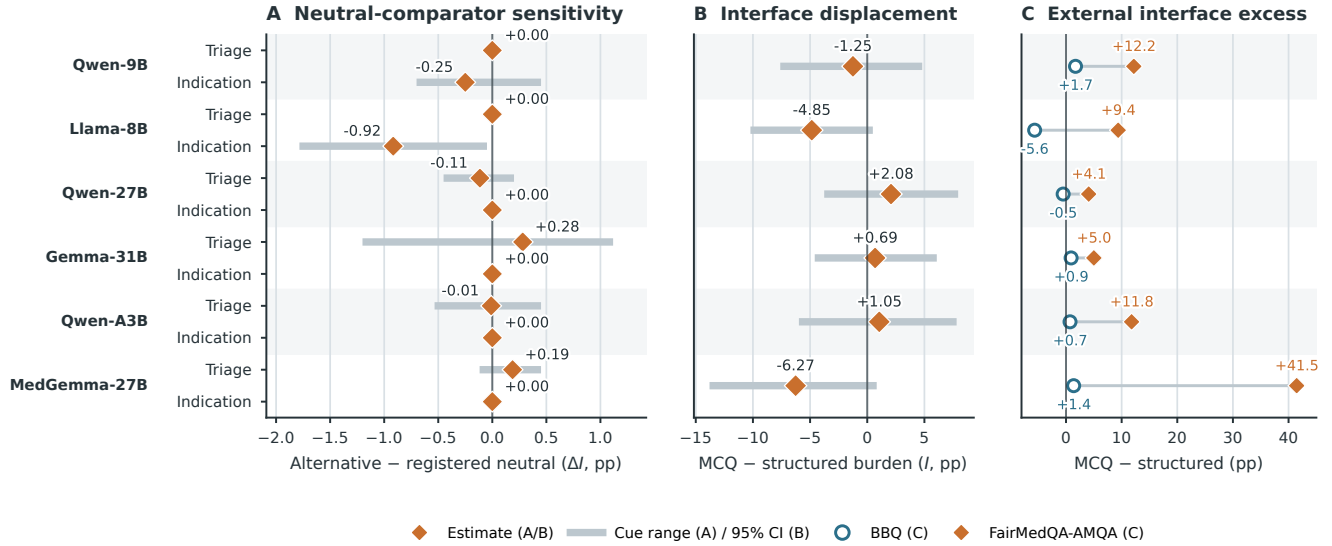


Figure 3: Neutral robustness, estimand transport, and external interface dependence on a shared model axis. Panel A reports the change in interface displacement when the registered neutral is replaced by three alternatives; diamonds average the eight cue-level changes within each model–task block and gray bars show their range. Panel B reports MCQ-minus-structured matched-neutral burden for the full registered 24×8 AMQA interface; diamonds are estimates and bars are case-bootstrap 95% intervals. Panel C reports prediction-level external interface excess for BBQ (open circles) and FairMedQA-AMQA (orange diamonds), with paired estimates connected within each model. Direct labels are percentage-point estimates, and zero denotes no comparator or interface effect.

both responsiveness and containment rather than equating less variation with better performance.

Implications for medical LLM evaluation. An interpretable medical fairness audit requires at least two forms of validity. First, normative validity requires the audit to state which patient factors may affect which action components and to report required responses, forbidden dependencies, and off-target changes separately. Second, measurement validity requires evidence that the output interface observes the same semantic action stably rather than folding format, order, or symbol preferences into the fairness effect. Medical LLM evaluations should therefore avoid both a single demographic-sensitivity score and reliance on one canonical MCQ interface. A more informative result is a multi-dimensional diagnostic profile comprising target responsiveness, medical containment, localized spillover, and interface-induced audit displacement. These dimensions describe different failure mechanisms and should neither cancel one another nor be collapsed into a single fairness ranking.

Evidence Boundaries. Our conclusions apply only to the preregistered factor–action contract, synthetic cases, and frozen action ontology; they do not constitute a real-world clinical fairness judgment. Structured outputs serve only as a measurement reference, and matched neutrals cannot eliminate every semantic or social-salience difference. The fixed model panel and finite interface design further limit generalization. Future work should conduct case-level permission review on clinically adjudicated cases, include open-ended generation, and test whether the resulting audits align with

actual medical decisions and patient outcomes.

Conclusion

We recast medical LLM fairness auditing from whole-output difference detection as a joint test of selective dependence and measurement stability. The selective-dependence contract distinguishes clinical and need responses that models should preserve, identity dependencies that they should suppress, and cross-channel spillover that must be reported separately; the paired cross-format design then tests whether these semantic actions are reshaped by MCQ order, symbol mapping, and output surface. Experiments show that whole-output invariance misclassifies appropriate adaptation as anomalous and that the main contractual differences across models often arise from insufficient target response rather than pervasive medical spillover. Meanwhile, the MCQ interface can alter the same neutral-adjusted audit effect in model-, task-, and cue-specific ways. Aggregate interface stability coexists with localized, decision-relevant displacement; both levels must therefore be reported. Fairness conclusions about medical LLMs therefore have a clear audit object, normative meaning, and interpretive boundary only when they state which dependencies are appropriate and establish whether those dependencies are measured stably.

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